

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I S.Harichandana(192372136)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:S.Harichandana

Annexure A

Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$$p = 17, q = 19$$

Step 1: Compute n

$$\text{Formula: } n = p \times q$$

$$n = 17 \times 19 = 323$$

Step 2: Compute Euler Totient $\phi(n)$

$$\text{Formula: } \phi(n) = (p-1)(q-1)$$

$$= (17-1)(19-1) = 16 \times 18 = 288$$

Step 3: Verify e

$$\gcd(5, 288) = 1,$$

so $e=5$ is valid.

Step 4: Find private key d

$$\text{Need: } 5d \equiv 1 \pmod{288}$$

$$\text{Try } d = 173.$$

$$5 \times 173 = 865$$

$$865 = 288 \times 3 + 1$$

$$\text{Remainder} = 1, \text{ therefore } d = 173.$$

$$\text{Public Key} = (5, 323)$$

$$\text{Private Key} = (173, 323)$$

Step 5: Encrypt

$$C = M^e \pmod{n} = 7^5 \pmod{323}$$

$$7^2 = 49$$

$$7^4 = 49 \times 49 = 2401$$

$$2401 = 323 \times 7 + 140 \Rightarrow 7^4 \equiv 140$$

$$7^5 = 140 \times 7 = 980$$

$$980 = 323 \times 3 + 11$$

$$\text{Ciphertext } C = 11$$

Step 6: Decrypt

$$M = C^d \pmod{n} = 11^{173} \pmod{323}$$

Using repeated modular exponentiation, result = 7.

2. ElGamal Encryption Analysis

Given:

$$p = 23, g = 5, \text{ private key } x = 7, \text{ random } k = 3, \text{ message } M = 10$$

Step 1: Compute public key**Formula:** $y = g^x \bmod p$

$$= 5^7 \bmod 23$$

$$5^2 = 25 \equiv 2$$

$$5^4 = 2^2 = 4$$

$$5^7 = 5^4 \times 5^2 \times 5 = 4 \times 2 \times 5 = 40$$

$$40 \bmod 23 = 17$$

$$\text{Public key} = (23, 5, 17)$$

Step 2: Compute C1

$$C1 = g^k \bmod p$$

$$p = 5^3 \bmod 23 = 125 \bmod 23 = 10$$

Step 3: Compute shared secret

$$y^k = 17^3 \bmod 23$$

$$17^2 = 289 = 23 \times 12 + 13$$

$$17^3 = 13 \times 17 = 221 = 23 \times 9 + 14$$

$$\text{Shared secret} = 14$$

Step 4: Compute C2

$$C2 = M \times \text{shared} \bmod p = 10 \times 14 = 140$$

$$140 = 23 \times 6 + 2$$

$$C2 = 2$$

$$\text{Ciphertext} = (10, 2)$$

Step 5: Decrypt

$$\text{Compute } C1^x = 10^7 \bmod 23 = 14$$

Find inverse of 14.

$$14 \times 5 = 70 = 23 \times 3 + 1$$

$$\text{Inverse} = 5$$

$$M = C2 \times \text{inverse} \bmod 23 = 2 \times 5 = 10 \bmod 23 = 10$$

Verification: Original message=10, Recovered message=10

Analysis:

Random value k changes every encryption.

Even the same message generates different ciphertexts.

This protects against pattern analysis and improves confidentiality.

3. RSA Digital Signature Verification**Given:**

$$p = 11, q = 17, e = 7, \text{ message } M = 8$$

Step 1: Compute n

$$n = p \times q = 11 \times 17 = 187$$

Step 2: Compute $\phi(n)$

$$\phi(n)=(11-1)(17-1)=10 \times 16=160$$

Step 3: Find Private Key d

$$\text{Need: } 7d \equiv 1 \pmod{160}$$

$$\text{Try } d=23$$

$$7 \times 23 = 161 = 160 \times 1 + 1$$

Therefore $d=23$.

Public Key=(7,187)

Private Key=(23,187)

Step 4: Generate Digital Signature

$$S = M^d \pmod{n}$$

$$S = 8^{23} \pmod{187}$$

Using repeated modular exponentiation,
Signature $S=57$.

Step 5: Verify Signature

$$M = S^e \pmod{n}$$

$$= 57^7 \pmod{187}$$

$$= 8$$

Verified message equals original message.

Analysis:

Only the private-key owner can generate the signature.

Verification with the public key proves authenticity, integrity and non-repudiation.

Question 4: Diffie-Hellman with Attack Resistance

Given: $p=29$, $g=2$

Private key of A=5

Private key of B=12

Step 1: Compute A's Public Key

$$A = g^a \pmod{p}$$

$$= 2^5 = 32$$

$$32 \pmod{29} = 3$$

Public Key of A=3

Step 2: Compute B's Public Key

$$B = g^b \pmod{p}$$

$$= 2^{12}$$

Public Key of B=7

Step 3: Compute Shared Secret

A computes:

$$7^5 \pmod{29} = 16$$

B computes:

$$3^{12} \bmod 29 = 16$$

Shared Secret Key=16

Attack Analysis:

If an attacker replaces public keys, both users unknowingly establish keys with the attacker (Man-in-the-Middle attack).

Solution:

Use Digital Certificates, RSA/ECDSA Digital Signatures or Authenticated Diffie-Hellman to authenticate public keys.

Question 5: Hybrid Encryption System Analysis

Given:

RSA key exchange time=5 ms

AES encryption time=0.5 ms/block

Total blocks=500

Step 1: RSA Time

=5 ms

Step 2: AES Time

= 500×0.5

=250 ms

Step 3: Total Hybrid Time

= $5 + 250$

=255 ms

Step 4: Only RSA

= 500×5

=2500 ms

Step 5: Comparison

Hybrid=255 ms

Only RSA=2500 ms

Time Saved= $2500 - 255 = 2245$ ms

Analysis:

RSA is slow but secure for key exchange.

AES is fast for bulk data encryption.

Using RSA only for key exchange and AES for data makes hybrid encryption highly efficient.

Justification:

Hybrid encryption is used in HTTPS, SSL/TLS, VPNs and secure messaging because it provides strong security with high performance.